

Infinite Horizon Models: Expected Discounted Reward Criterion

MDP:

$$S_0 \xrightarrow{A_0} (S_1, R_1) \xrightarrow{A_1} (S_2, R_2) \xrightarrow{A_2} \dots$$

no finite horizon

In general,

$$R_n = r_n(S_{n-1}, A_{n-1}, S_n)$$

Key Assumptions:

We will assume the following:

- ① State space $\mathcal{S}$ is finite.
- ② Action space $\mathcal{A}$ is finite.
- ③ There exists $r: \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [-m, m]$ such that

bounded and stationary rewards

$$R_n = r(S_{n-1}, A_{n-1}, S_n) \quad \forall n \in \mathbb{N}.$$

- ④ TPM of the MDP, P , is time-homogeneous.
- ⑤ Discount factor $\lambda \in [0, 1]$.

Shorthand representation: $M = (\mathcal{S}, \mathcal{A}, r, P, \lambda)$

all these are assumed to be known

We introduce a new notation:

$$r(s, a) := \sum_{s' \in \mathcal{S}} P(s'|s, a) r(s, a, s') = \mathbb{E}^\pi [r(S_{n-1}, A_{n-1}, S_n) | S_{n-1} = s, A_{n-1} = a]$$

one-step expected reward obtained by taking action a in state s

Recall the definition of $V_{\infty, \lambda}^\pi(s)$:

$$\begin{aligned} V_{\infty, \lambda}^\pi(s) &:= \lim_{N \rightarrow \infty} \mathbb{E}^\pi \left[\sum_{n=1}^N \lambda^{n-1} R_n \mid S_0 = s \right] \\ &= \lim_{N \rightarrow \infty} \mathbb{E}^\pi \left[\sum_{n=1}^N \lambda^{n-1} r(S_{n-1}, A_{n-1}) \mid S_0 = s \right] \\ &\stackrel{\text{DCT}}{=} \mathbb{E}^\pi \left[\sum_{n=1}^{\infty} \lambda^{n-1} r(S_{n-1}, A_{n-1}) \mid S_0 = s \right] \end{aligned}$$

$\mathbb{E}^\pi [R_n | S_0 = s] = \mathbb{E}^\pi [\mathbb{E}^\pi [R_n | S_{n-1}, A_{n-1}] | S_0 = s]$

$\pi \in \Pi^{\text{MR}} \Rightarrow \{S_n\}_{n \geq 0}$ is a DTMC

$$P^\pi(S_{n+1}=s' | S_n=s) = \sum_{a \in \mathcal{A}} P^\pi(S_{n+1}=s', A_n=a | S_n=s)$$

$$= \sum_{a \in \mathcal{A}} \pi_n(a|s) \cdot P(s'|s, a).$$

$$= P_{\pi_n}(s'|s).$$

$\pi \in \Pi^{\text{SR}} \Rightarrow \pi = \{\phi, \phi, \phi, \dots\}$, $\phi: \mathcal{S} \rightarrow \Delta(\mathcal{A})$

$$P^\pi(S_{n+1}=s' | S_n=s) = \sum_{a \in \mathcal{A}} \phi(a|s) P(s'|s, a)$$

$$= P_\phi(s'|s).$$

$$\stackrel{\text{DCT}}{=} \sum_{n=1}^{\infty} \lambda^{n-1} \mathbb{E}^{\pi} [r(S_{n-1}, A_{n-1}) | S_0 = s].$$

Theorem 5.1: Given any $\pi = \{\pi_n\}_{n \geq 1} \in \Pi^{\text{MR}}$, there exists $\pi' = \{\pi'_n\}_{n \geq 1} \in \Pi^{\text{MR}}$ such that

$$\sum_{n=1}^{\infty} \lambda^{n-1} \mathbb{E}^{\pi} [r(S_{n-1}, A_{n-1}) | S_0 = s] = \sum_{n=1}^{\infty} \lambda^{n-1} \mathbb{E}^{\pi'} [r(S_{n-1}, A_{n-1}) | S_0 = s] \text{ for all } s \in \mathcal{S}.$$

(Proof deferred until later)

Observe that

$$V_{\infty, \lambda}^{\pi}(s) = \sum_{n=1}^{\infty} \lambda^{n-1} \underbrace{\sum_{s' \in \mathcal{S}} \sum_{a' \in \mathcal{A}} r(s', a') \cdot \mathbb{P}^{\pi}(S_{n-1} = s', A_{n-1} = a' | S_0 = s)}$$

Assuming that $\pi \in \Pi^{\text{MR}}$ (sufficient to look at MR policies in light of Theorem 5.1):

$$\underline{n=1}: \sum_{a' \in \mathcal{A}} r(s, a') \mathbb{P}^{\pi}(A_0 = a' | S_0 = s) = \sum_{a' \in \mathcal{A}} r(s, a') \pi_0(a' | s).$$

$$\underline{n=2}: \lambda \sum_{s' \in \mathcal{S}} \sum_{a' \in \mathcal{A}} r(s', a') \mathbb{P}^{\pi}(S_1 = s', A_1 = a' | S_0 = s) \stackrel{:= r_{\pi_0}(s)}{\leftarrow \text{expected reward going out of state } s \text{ at time } n=0}$$

$$= \lambda \sum_{s' \in \mathcal{S}} \sum_{a' \in \mathcal{A}} r(s', a') \cdot \sum_{a \in \mathcal{A}} \mathbb{P}^{\pi}(S_1 = s', A_1 = a', A_0 = a | S_0 = s)$$

$$= \lambda \sum_{s' \in \mathcal{S}} \sum_{a' \in \mathcal{A}} r(s', a') \cdot \sum_{a \in \mathcal{A}} \pi_0(a | s) \cdot P(s' | s, a) \cdot \pi_1(a' | s')$$

$$= \lambda \sum_{s' \in \mathcal{S}} \sum_{a' \in \mathcal{A}} r(s', a') \pi_1(a' | s') \cdot \sum_{a \in \mathcal{A}} P(s' | s, a) \pi_0(a | s) \stackrel{:= r_{\pi_1}(s')}{=} \stackrel{:= P_{\pi_0}(s' | s)}{=}$$

and so on...

expected reward going out of state s' at time $n=1$

Notation:

Recall that if $\pi \in \Pi^{\text{MR}}$, then $\{S_n\}_{n \geq 0}$ generated by π is a DTMC. We shall denote its TPM by

$$P_{\pi} = \{P_{\pi_0}, P_{\pi_1}, P_{\pi_2}, \dots\}$$

$$\underline{n=1}: r_{\pi_0}(s) := \sum_{a \in \mathcal{A}} r(s, a) \pi_0(a|s) = r(s, \cdot)^T \pi_0(\cdot|s)$$

$$\underline{n=2}: \lambda \sum_{s' \in \mathcal{S}} r_{\pi_1}(s') \frac{P_{\pi_0}(s'|s)}{P_{\pi_1}(s'|s)}$$

$$\underline{n=3}: \lambda^2 \sum_{s' \in \mathcal{S}} \sum_{s'' \in \mathcal{S}} r_{\pi_2}(s'') \frac{P_{\pi_1}(s''|s')}{P_{\pi_0}(s'|s)} \quad (\text{exercise})$$

$$= \lambda^2 \sum_{s'' \in \mathcal{S}} r_{\pi_2}(s'') \cdot \sum_{s' \in \mathcal{S}} \frac{P_{\pi_1}(s''|s')}{P_{\pi_0}(s'|s)} \cdot P_{\pi_0}(s'|s)$$

Because π is a Markov policy, we know that $\{S_n\}_{n \geq 0}$ is a time-homogeneous DTMC under π , and hence

$$\sum_{s' \in \mathcal{S}} \frac{P_{\pi_1}(s''|s')}{P_{\pi_0}(s'|s)} \cdot P_{\pi_0}(s'|s) = P^\pi(S_2 = s'' | S_0 = s)$$

represents the 2-step state-transition probability from s to s''

Going forward, we will write

$$\frac{P_\pi^2}{\pi}(s''|s) := \sum_{s' \in \mathcal{S}} \frac{P_{\pi_1}(s''|s')}{P_{\pi_0}(s'|s)} \cdot P_{\pi_0}(s'|s)$$

In matrix-vector notation:

$$\underline{n=1}: r_{\pi_0} = \lambda^0 \frac{P_\pi^0}{\pi} r_{\pi_0}$$

$$P_\pi^n = P_{\pi_0} P_{\pi_1} \dots P_{\pi_{n-1}}$$

$$\underline{n=2}: \lambda \frac{P_\pi^1}{\pi} r_{\pi_1} = \lambda \frac{P_\pi^1}{\pi} r_{\pi_1}$$

$$\underline{n=3}: \lambda^2 \frac{P_\pi^2}{\pi} r_{\pi_2} = \lambda^2 \frac{P_\pi^2}{\pi} r_{\pi_2}$$

:

Hence,

$$V_{\infty, \lambda}^\pi = \sum_{n=1}^{\infty} \lambda^{n-1} \frac{P_\pi^{n-1}}{\pi} r_{\pi_{n-1}}$$

Is this infinite sum defined? Yes, because the tail sum

$$\begin{aligned} \left\| \sum_{n=N}^{\infty} \lambda^{n-1} \frac{P_\pi^{n-1}}{\pi} r_{\pi_{n-1}} \right\|_{\infty} &\leq \sum_{n=N}^{\infty} \lambda^{n-1} \left\| \frac{P_\pi^{n-1}}{\pi} r_{\pi_{n-1}} \right\|_{\infty} \\ &\leq \sum_{n=N}^{\infty} \lambda^{n-1} \left\| \frac{P_\pi^{n-1}}{\pi} \right\| \left\| r_{\pi_{n-1}} \right\|_{\infty} \end{aligned}$$

Sub-norm

↑
matrix norm

$$\|Q\| = \sup_{\|v\|_\infty = 1} \|Qv\|_\infty = \max_i \sum_j |Q_{ij}|$$

bounded rewards

$$\|r_{\pi_{n-1}}\|_\infty \leq M \leq \sum_{n=N}^{\infty} \lambda^{n-1} \cdot M \quad \left(\|P_\pi^{n-1}\| = 1 \right)$$

$$= M \frac{\lambda^{N-1}}{1-\lambda}$$

$$\rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

Continuing, we observe that

$$V_{\infty, \lambda}^\pi = r_{\pi_0} + \lambda P_\pi^1 r_{\pi_1} + \lambda^2 P_\pi^2 r_{\pi_2} + \dots$$

$$= r_{\pi_0} + \lambda P_{\pi_0} r_{\pi_1} + \lambda^2 P_{\pi_0} P_{\pi_1} r_{\pi_2} + \dots$$

$$= r_{\pi_0} + \lambda P_{\pi_0} \left(r_{\pi_1} + \lambda P_{\pi_1} r_{\pi_2} + \dots \right)$$

$$= r_{\pi_0} + \lambda P_{\pi_0} V_{\infty, \lambda}^{\pi'},$$

where $\pi' = \{\pi_1, \pi_2, \pi_3, \dots\}$ is a shifted version of π (without π_0).

Summary so far:

$$V_{\infty, \lambda}^\pi = r_{\pi_0} + \lambda P_{\pi_0} V_{\infty, \lambda}^{\pi'}$$

$$V_{\infty, \lambda}^\pi(s) = \sum_{a' \in \mathcal{A}} \pi_0(a'|s) r(a', s) + \lambda \sum_{s' \in \mathcal{S}} P_{\pi_0}(s'|s) V_{\infty, \lambda}^{\pi'}(s')$$

$$V_{\infty, \lambda}^\pi(s) = \mathbb{E}^\pi \left[r(S_0, A_0) + \lambda V_{\infty, \lambda}^{\pi'}(S_1) \mid S_0 = s \right]$$

Evaluating $V_{\infty, \lambda}^{\pi}$ for $\pi \in \Pi^{SR}$

Let $\pi = \{\phi, \phi, \phi, \dots\}$ be a stationary, randomised policy, where

$$\phi: \mathcal{S} \rightarrow \Delta(\mathcal{A})$$

is a fixed, known function.

- Under $\pi = \{\phi, \phi, \dots\}$, the process $\{S_n\}_{n \geq 0}$ is a time-homogeneous DTMC with TPM

$$P_{\phi}(s'|s) = \sum_{a \in \mathcal{A}} \phi(a|s) P(s'|s, a) \quad \forall s, s' \in \mathcal{S}.$$

and expected reward in state s

$$r_{\phi}(s) = \sum_{a \in \mathcal{A}} \phi(a|s) r(s, a)$$

- In this case,

$$\begin{aligned} V_{\infty, \lambda}^{\pi} &= V_{\infty, \lambda}^{\phi} = \sum_{n=1}^{\infty} \lambda^{n-1} P_{\phi}^{n-1} r_{\phi} \\ &= r_{\phi} + \lambda P_{\phi} V_{\infty, \lambda}^{\phi} \end{aligned}$$

- Thus, $V_{\infty, \lambda}^{\pi} = V_{\infty, \lambda}^{\phi}$ satisfies

$$V_{\infty, \lambda}^{\phi} = r_{\phi} + \lambda P_{\phi} V_{\infty, \lambda}^{\phi}$$

- Equivalently, $V_{\infty, \lambda}^{\phi}$ is the solution to the fixed point equation

$$V = r_{\phi} + \lambda P_{\phi} V, \quad V \in [-M, M]^{|S|}$$

$$\Leftrightarrow (I - \lambda P_{\phi}) V = r_{\phi}.$$

Lemma 5.2: Suppose $Q = [Q_{ij}]$ is an $|S| \times |S|$ real matrix satisfying

$$Q^n \xrightarrow{n \rightarrow \infty} 0 \quad \text{entry-wise.}$$

$|S| \times |S|$ matrix of zero entries

Then, $(I - Q)$ is invertible, and

$$(I - Q)^{-1} = I + Q + Q^2 + \dots = \sum_{n=0}^{\infty} Q^n.$$

Proof of Lemma 5.2:

For any $n \geq 0$, let

$$S_n = \sum_{i=0}^n Q^i$$

($Q^0 = I$ by convention)

Then, we have

$$(I - Q) S_{n-1} = I - Q^n \quad \forall n \geq 1.$$

Taking limits as $n \rightarrow \infty$, and using $Q^n \rightarrow 0$, we get

$$\lim_{n \rightarrow \infty} (I - Q) S_{n-1} = I$$

$$\Leftrightarrow (I - Q) \lim_{n \rightarrow \infty} S_{n-1} = I$$

$$\Leftrightarrow (I - Q) \left(\sum_{n=0}^{\infty} Q^n \right) = I,$$

from which the result follows.

Observe that if $\lambda \in [0, 1)$, then

$$(\lambda P_\phi)^n \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Applying Lemma 5.2 to $Q = \lambda P_\phi$, we get that $(I - \lambda P_\phi)$ is invertible, and hence the equation

$$(I - \lambda P_\phi) v = r_\phi$$

has a unique solution given by

$$v^* = (I - \lambda P_\phi)^{-1} r_\phi.$$

Theorem 5.2 : Suppose $\lambda \in [0, 1)$ and $\pi = \{\phi, \phi, \dots\}$. Then,

$$v_{\infty, \lambda}^\phi = (I - \lambda P_\phi)^{-1} r_\phi$$

Some properties of $(I - \lambda P_\phi)^{-1}$

① If $\underline{u} \geq \underline{0}$ component-wise, then

$$(I - \lambda P_\phi)^{-1} \underline{u} \geq \underline{u} \geq \underline{0}.$$

② If $\underline{u} \geq \underline{v}$ component-wise, then

$$(I - \lambda P_\phi)^{-1} \underline{u} \geq (I - \lambda P_\phi)^{-1} \underline{v}.$$

$$\textcircled{3} \quad (I - \lambda P_\phi)^{-1} \underline{1} = \frac{1}{1-\lambda} \underline{1}.$$

Proof: From Lemma 5.2, we know

$$(I - \lambda P_\phi)^{-1} = \sum_{n=0}^{\infty} \lambda^n P_\phi^n.$$

Then, part ① follows by noting that

$$(I - \lambda P_\phi)^{-1} \underline{u} = \underline{u} + \lambda P_\phi \underline{u} + \lambda^2 P_\phi^2 \underline{u} + \dots$$

$$\geq \underline{u} \quad \left(P_\phi^n \underline{u} \geq \underline{0} \text{ component-wise for all } n \right)$$

Part ② follows by applying part ① to

$$\underline{u} \leftarrow \underline{u} - \underline{v}.$$

Part ③ is a simple consequence of

$$P_\phi^n \underline{1} = \underline{1} \quad \forall n \geq 0,$$

$$\sum_{n=0}^{\infty} \lambda^n = \frac{1}{1-\lambda}, \quad \lambda \in [0, 1).$$
